PROBABILITY

1.	If $P\left(\frac{A}{B}\right) = 0{\cdot}3$, $P(A) = 0{\cdot}4$ and $P(B) = 0{\cdot}8$, then $P\left(\frac{B}{A}\right)$ is equal to : (a) 0·6 (b) 0·3 (c) 0·06 (d) 0·4
2.	The probability distribution of a random variable X is given below : <table><tr><td>X</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X)</td><td>$\frac{k}{2}$</td><td>$\frac{k}{3}$</td><td>$\frac{k}{6}$</td></tr></table>(i) Find the value of k. (ii) Find $P(1 \leq X < 3)$. (iii) Find E(X), the mean of X.
3.	A and B are independent events such that $P(A \cap \bar{B}) = \frac{1}{4}$ and $P(\bar{A} \cap B) = \frac{1}{6}$. Find P(A) and P(B).
4.	If A and B are two events such that $P(A/B) = 2 \times P(B/A)$ and $P(A) + P(B) = \frac{2}{3}$, then P(B) is equal to (A) $\frac{2}{9}$ (B) $\frac{7}{9}$ (C) $\frac{4}{9}$ (D) $\frac{5}{9}$
5.	Two balls are drawn at random one by one with replacement from an urn containing equal number of red balls and green balls. Find the probability distribution of number of red balls. Also, find the mean of the random variable.
6.	X and Y are independent events such that $P(X \cap \bar{Y}) = \frac{2}{5}$ and $P(X) = \frac{3}{5}$. Then P(Y) is equal to : (a) $\frac{2}{3}$ (b) $\frac{2}{5}$ (c) $\frac{1}{3}$ (d) $\frac{1}{5}$

7.	A and B throw a die alternately till one of them gets a '6' and wins the game. Find their respective probabilities of wining, if A starts the game first.
8.	A pair of dice is thrown simultaneously. If X denotes the absolute difference of numbers obtained on the pair of dice, then find the probability distribution of X.
9.	There are two coins. One of them is a biased coin such that P (head) : P (tail) is 1 : 3 and the other coin is a fair coin. A coin is selected at random and tossed once. If the coin showed head, then find the probability that it is a biased coin.
10.	The probability that A speaks the truth is $\frac{4}{5}$ and that of B speaking the truth is $\frac{3}{4}$. The probability that they contradict each other in stating the same fact is : (a) $\frac{7}{20}$ (b) $\frac{1}{5}$ (c) $\frac{3}{20}$ (d) $\frac{4}{5}$
11.	From a lot of 30 bulbs which include 6 defective bulbs, a sample of 2 bulbs is drawn at random one by one with replacement. Find the probability distribution of the number of defective bulbs and hence find the mean number of defective bulbs.
12.	If for any two events A and B, $P(A) = \frac{4}{5}$ and $P(A \cap B) = \frac{7}{10}$, then P(B/A) is equal to (a) $\frac{1}{10}$ (b) $\frac{1}{8}$ (c) $\frac{7}{8}$ (d) $\frac{17}{20}$
13.	If A and B are two independent events such that $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{4}$, then $P(B'/A)$ is (a) $\frac{1}{4}$ (b) $\frac{1}{8}$ (c) $\frac{3}{4}$ (d) 1

14.	**Assertion (A) :** Two coins are tossed simultaneously. The probability of getting two heads, if it is known that at least one head comes up, is $\frac{1}{3}$. **Reason (R) :** Let E and F be two events with a random experiment, then $P(F/E) = \frac{P(E \cap F)}{P(E)}$.
15.	In answering a question on a multiple choice test, a student either knows the answer or guesses. Let $\frac{3}{5}$ be the probability that he knows the answer and $\frac{2}{5}$ be the probability that he guesses. Assuming that a student who guesses at the answer will be correct with probability $\frac{1}{3}$. What is the probability that the student knows the answer, given that he answered it correctly ?
16.	A box contains 10 tickets, 2 of which carry a prize of ₹ 8 each, 5 of which carry a prize of ₹ 4 each, and remaining 3 carry a prize of ₹ 2 each. If one ticket is drawn at random, find the mean value of the prize.
17.	*Assertion (A) :* If R and S are two events such that $P(R \mid S) = 1$ and $P(S) > 0$, then $S \subset R$. *Reason (R) :* If two events A and B are such that $P(A \cap B) = P(B)$, then $A \subset B$.
18.	It is known that 20% of the students in a school have above 90% attendance and 80% of the students are irregular. Past year results show that 80% of students who have above 90% attendance and 20% of irregular students get 'A' grade in their annual examination. At the end of a year, a student is chosen at random from the school and is found to have an 'A' grade. What is the probability that the student is irregular ?
19.	A family has 2 children and the elder child is a girl. The probability that both children are girls is : (a) $\frac{1}{4}$ (b) $\frac{1}{8}$ (c) $\frac{1}{2}$ (d) $\frac{3}{4}$

20.	Out of two bags, bag A contains 2 white and 3 red balls and bag B contains 4 white and 5 red balls. One ball is drawn at random from one of the bags and is found to be red. Find the probability that it was drawn from bag B.
21.	Out of a group of 50 people, 20 always speak the truth. Two persons are selected at random from the group (without replacement). Find the probability distribution of number of selected persons who always speak the truth.
22.	If $P(A \mid B) = P(A' \mid B)$, then which of the following statements is true ? (A) $P(A) = P(A')$ (B) $P(A) = 2\,P(B)$ (C) $P(A \cap B) = \frac{1}{2}\,P(B)$ (D) $P(A \cap B) = 2\,P(B)$
23.	E and F are two independent events such that $P(\bar{E}) = 0{\cdot}6$ and $P(E \cup F) = 0{\cdot}6$. Find $P(F)$ and $P(\bar{E} \cup \bar{F})$.
24.	Let E be an event of a sample space S of an experiment, then $P(S \mid E) =$ (A) $P(S \cap E)$ (B) $P(E)$ (C) 1 (D) 0
25.	A pair of dice is thrown simultaneously. If X denotes the absolute difference of the numbers appearing on top of the dice, then find the probability distribution of X.
26.	Let E and F be two events such that $P(E) = 0{\cdot}1$, $P(F) = 0{\cdot}3$, $P(E \cup F) = 0{\cdot}4$, then $P(F \mid E)$ is : (A) $0{\cdot}6$ (B) $0{\cdot}4$ (C) $0{\cdot}5$ (D) 0
27.	The chances of P, Q and R getting selected as CEO of a company are in the ratio 4 : 1 : 2 respectively. The probabilities for the company to increase its profits from the previous year under the new CEO, P, Q or R are $0{\cdot}3$, $0{\cdot}8$ and $0{\cdot}5$ respectively. If the company increased the profits from the previous year, find the probability that it is due to the appointment of R as CEO.

28.	The probability distribution of a random variable X is :

X	0	1	2	3	4
P(X)	0.1	k	2k	k	0.1

where k is some unknown constant.

The probability that the random variable X takes the value 2 is :

(A) $\frac{1}{5}$ (B) $\frac{2}{5}$

(C) $\frac{4}{5}$ (D) 1

29. A card from a well shuffled deck of 52 playing cards is lost. From the remaining cards of the pack, a card is drawn at random and is found to be a King. Find the probability of the lost card being a King.

30. A biased die is twice as likely to show an even number as an odd number. If such a die is thrown twice, find the probability distribution of the number of sixes. Also, find the mean of the distribution.

31. If A and B are events such that $P(A/B) = P(B/A) \neq 0$, then :

(A) $A \subset B$, but $A \neq B$ (B) $A = B$

(C) $A \cap B = \phi$ (D) $P(A) = P(B)$

32. The random variable X has the following probability distribution where a and b are some constants :

X	1	2	3	4	5
P(X)	0.2	a	a	0.2	b

If the mean $E(X) = 3$, then find values of a and b and hence determine $P(X \geq 3)$.

33. If E and F are two independent events such that $P(E) = \frac{2}{3}$, $P(F) = \frac{3}{7}$, then $P(E/\overline{F})$ is equal to :

(A) $\frac{1}{6}$ (B) $\frac{1}{2}$

(C) $\frac{2}{3}$ (D) $\frac{7}{9}$

34. The probability distribution for the number of students being absent in a class on a Saturday is as follows :

X	0	2	4	5
P(X)	p	2p	3p	p

Where X is the number of students absent.

(i) Calculate p.

(ii) Calculate the mean of the number of absent students on Saturday.

35. For the vacancy advertised in the newspaper, 3000 candidates submitted their applications. From the data it was revealed that two third of the total applicants were females and other were males. The selection for the job was done through a written test. The performance of the applicants indicates that the probability of a male getting a distinction in written test is 0.4 and that a female getting a distinction is 0.35. Find the probability that the candidate chosen at random will have a distinction in the written test.

36. In the following probability distribution, the value of p is :

X	0	1	2	3
P(X)	p	p	0.3	2p

(A) $\frac{7}{40}$ (B) $\frac{1}{10}$

(C) $\frac{9}{35}$ (D) $\frac{1}{4}$

37.	If E and F are two events such that $P(E) > 0$ and $P(F) \neq 1$, then $P(\overline{E}/\overline{F})$ is (A) $\dfrac{P(\overline{E})}{P(\overline{F})}$ (B) $1 - P(\overline{E}/F)$ (C) $1 - P(E/F)$ (D) $\dfrac{1 - P(E \cup F)}{P(\overline{F})}$
38.	A die with number 1 to 6 is biased such that $P(2) = \dfrac{3}{10}$ and probability of other numbers is equal. Find the mean of the number of times number 2 appears on the dice, if the dice is thrown twice.
39.	Two dice are thrown. Defined are the following two events A and B : $A = \{(x, y) : x + y = 9\}$, $B = \{(x, y) : x \neq 3\}$, where (x, y) denote a point in the sample space. Check if events A and B are independent or mutually exclusive.
40.	*Assertion (A) :* If A and B are two events such that $P(A \cap B) = 0$, then A and B are independent events. *Reason (R) :* Two events are independent if the occurrence of one does not effect the occurrence of the other.
41.	The probability distribution of a random variable X is given by : X: 0, 1, 2, 3 P(X): p, $\frac{p}{3}$, $\frac{p}{6}$, $\frac{p}{12}$ (i) Determine the value of p. (ii) Calculate $P(X \geq 1)$. (iii) Calculate expectation of X, i.e. E(X)

Table for Q41:

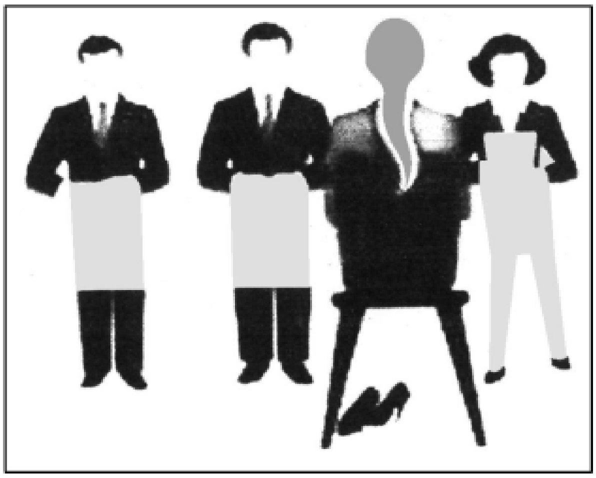
X	0	1	2	3
P(X)	p	$\frac{p}{3}$	$\frac{p}{6}$	$\frac{p}{12}$

42.	In a city, a survey was conducted among residents about their preferred mode of commuting. It was found that 50% people preferred using public transport, 35% preferred using a bicycle and 20% use both public transport and a bicycle. If a person is selected at random, find the probability that : (i) The person uses only public transport. *1* (ii) The person uses a bicycle, given that they also use the public transport. *1* (iii) The person uses neither public transport nor a bicycle. *1*
43.	If $P(A \cup B) = 0{\cdot}9$ and $P(A \cap B) = 0{\cdot}4$, then $P(\overline{A}) + P(\overline{B})$ is : (A) 0·3 (B) 1 (C) 1·3 (D) 0·7
44.	The probability that a student buys a colouring book is 0·7 and that she buys a box of colours is 0·2. The probability that she buys a colouring book, given that she buys a box of colours, is 0·3. Find the probability that the student : (i) Buys both the colouring book and the box of colours. (ii) Buys a box of colours given that she buys the colouring book.
45.	A person has a fruit box that contains 6 apples and 4 oranges. He picks out a fruit three times, one after the other, after replacing the previous one in the box. Find : (i) The probability distribution of the number of oranges he draws. (ii) The expectation of the random variable (number of oranges).

46.	Chances that three persons A, B, and C go to the market are 30%, 60% and 50% respectively. The probability that at least one will go to the market is : (A) $\frac{14}{10}$ (B) $\frac{43}{50}$ (C) $\frac{9}{100}$ (D) $\frac{7}{50}$
47.	A coin is biased so that the head is 3 times as likely to occur as tail. If the coin is tossed three times, find the probability distribution of number of tails. Hence, find the mean of the distribution.
48.	A coin is tossed and a card is selected at random from a well shuffled pack of 52 playing cards. The probability of getting head on the coin and a face card from the pack is : (A) $\frac{2}{13}$ (B) $\frac{3}{26}$ (C) $\frac{19}{26}$ (D) $\frac{3}{13}$
49.	If A and B are two events such that $P(B) = \frac{1}{5}$, $P(A \mid B) = \frac{2}{3}$ and $P(A \cup B) = \frac{3}{5}$, then P(A) is : (A) $\frac{10}{15}$ (B) $\frac{2}{15}$ (C) $\frac{1}{5}$ (D) $\frac{8}{15}$
50.	10 identical blocks are marked with '0' on two of them, '1' on three of them, '2' on four of them and '3' on one of them and put in a box. If X denotes the number written on the block, then write the probability distribution of X and calculate its mean.
51.	In a village of 8000 people, 3000 go out of the village to work and 4000 are women. It is noted that 30% of women go out of the village to work. What is the probability that a randomly chosen individual is either a woman or a person working outside the village ?

	CASE STUDY
52.	A departmental store sends bills to charge its customers once a month. Past experience shows that 70% of its customers pay their first month bill in time. The store also found that the customer who pays the bill in time has the probability of 0.8 of paying in time next month and the customer who doesn't pay in time has the probability of 0.4 of paying in time the next month. Based on the above information, answer the following questions : (i) Let E_1 and E_2 respectively denote the event of customer paying or not paying the first month bill in time. Find $P(E_1)$, $P(E_2)$. (ii) Let A denotes the event of customer paying second month's bill in time, then find $P(A \mid E_1)$ and $P(A \mid E_2)$. (iii) Find the probability of customer paying second month's bill in time. **OR** (iii) Find the probability of customer paying first month's bill in time if it is found that customer has paid the second month's bill in time.
53.	A shop selling electronic items sells smartphones of only three reputed companies A, B and C because chances of their manufacturing a defective smartphone are only 5%, 4% and 2% respectively. In his inventory he has 25% smartphones from company A, 35% smartphones from company B and 40% smartphones from company C. A person buys a smartphone from this shop. (i) Find the probability that it was defective. *2* (ii) What is the probability that this defective smartphone was manufactured by company B ? *2*

54. Rohit, Jaspreet and Alia appeared for an interview for three vacancies in the same post. The probability of Rohit's selection is $\frac{1}{5}$, Jaspreet's selection is $\frac{1}{3}$ and Alia's selection is $\frac{1}{4}$. The event of selection is independent of each other.



Based on the above information, answer the following questions :

(i) What is the probability that at least one of them is selected ?

(ii) Find $P(G \mid \overline{H})$ where G is the event of Jaspreet's selection and $\overline{H}$ denotes the event that Rohit is not selected.

(iii) Find the probability that exactly one of them is selected.

OR

(iii) Find the probability that exactly two of them are selected.

55. Based upon the results of regular medical check-ups in a hospital, it was found that out of 1000 people, 700 were very healthy, 200 maintained average health and 100 had a poor health record.

Let A_1 : People with good health,

A_2 : People with average health,

and A_3 : People with poor health.

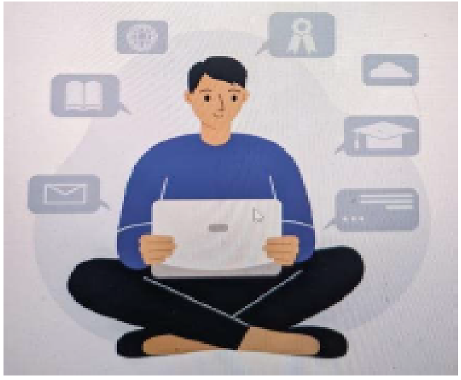
During a pandemic, the data expressed that the chances of people contracting the disease from category A_1, A_2 and A_3 are 25%, 35% and 50%, respectively.

Based upon the above information, answer the following questions :

(i) A person was tested randomly. What is the probability that he/she has contracted the disease ? *2*

(ii) Given that the person has not contracted the disease, what is the probability that the person is from category A_2 ? *2*

56. Self-study helps students to build confidence in learning. It boosts the self-esteem of the learners. Recent surveys suggested that close to 50% learners were self-taught using internet resources and upskilled themselves.



A student may spend 1 hour to 6 hours in a day in upskilling self. The probability distribution of the number of hours spent by a student is given below :

$$P(X = x) = \begin{cases} kx^2, & \text{for } x = 1, 2, 3 \\ 2kx, & \text{for } x = 4, 5, 6 \\ 0, & \text{otherwise} \end{cases}$$

where x denotes the number of hours.

Based on the above information, answer the following questions :

(i) Express the probability distribution given above in the form of a probability distribution table.

(ii) Find the value of k.

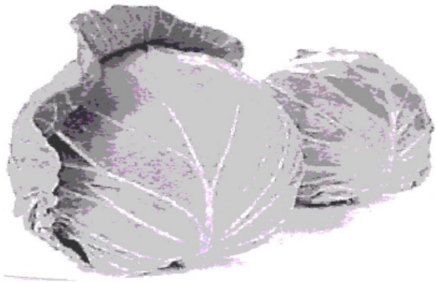
(iii) (a) Find the mean number of hours spent by the student.

OR

(iii) (b) Find $P(1 < X < 6)$.

57. A gardener wanted to plant vegetables in his garden. Hence he bought 10 seeds of brinjal plant, 12 seeds of cabbage plant and 8 seeds of radish plant. The shopkeeper assured him of germination probabilities of brinjal, cabbage and radish to be 25%, 35% and 40% respectively. But before he could plant the seeds, they got mixed up in the bag and he had to sow them randomly.

Radish



Cabbage

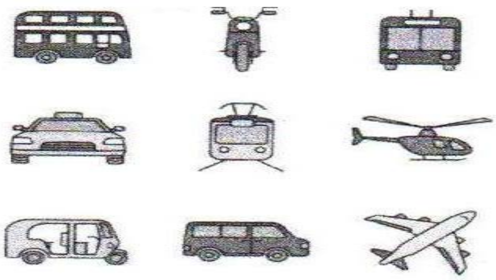
Brinjal

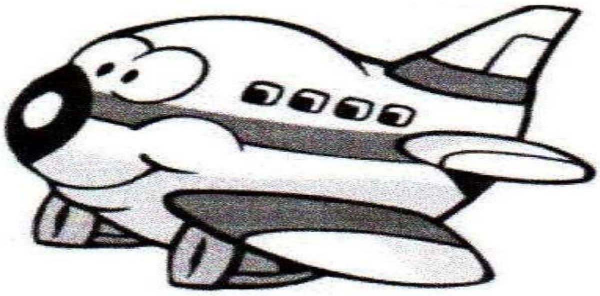
Based upon the above information, answer the following questions :

(i) Calculate the probability of a randomly chosen seed to germinate. *2*

(ii) What is the probability that it is a cabbage seed, given that the chosen seed germinates ? *2*

58. Airplanes are by far the safest mode of transportation when the number of transported passengers are measured against personal injuries and fatality totals.





Previous records state that the probability of an airplane crash is 0·00001%. Further, there are 95% chances that there will be survivors after a plane crash. Assume that in case of no crash, all travellers survive.

Let E_1 be the event that there is a plane crash and E_2 be the event that there is no crash. Let A be the event that passengers survive after the journey.

On the basis of the above information, answer the following questions :

(i) Find the probability that the airplane will not crash.

(ii) Find $P(A \mid E_1) + P(A \mid E_2)$.

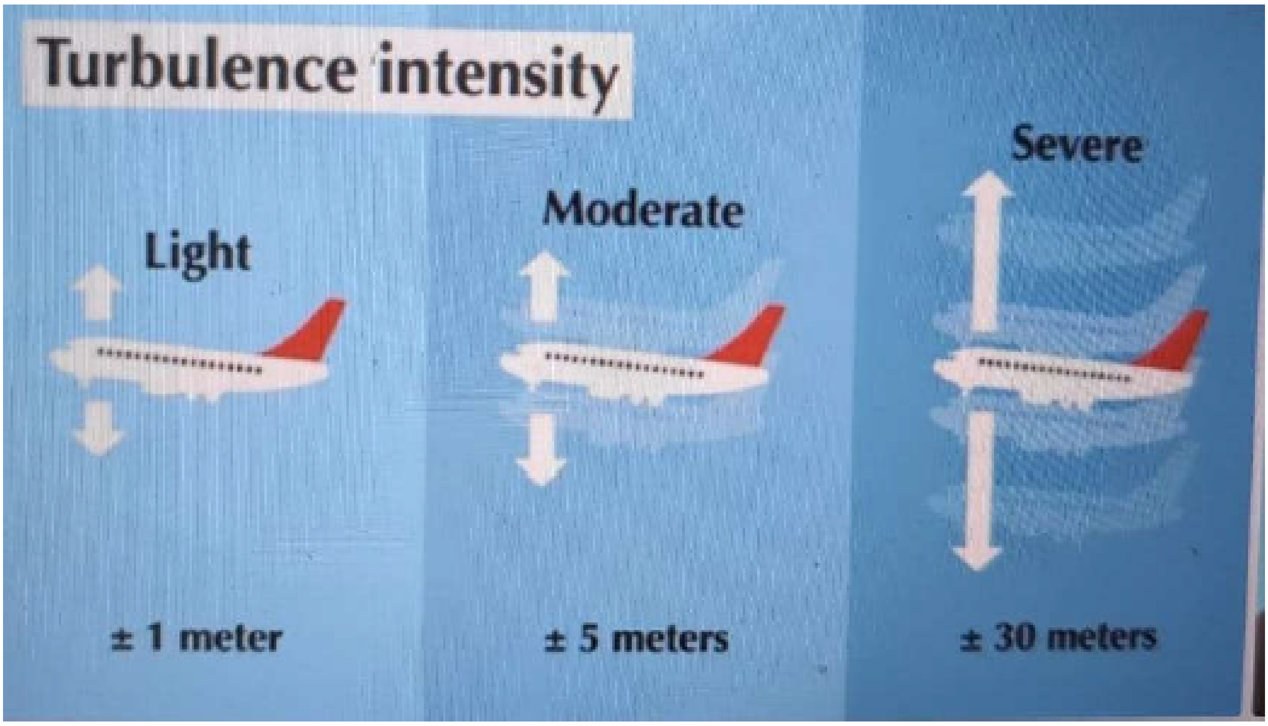
(iii) (a) Find P(A).

OR

(iii) (b) Find $P(E_2 \mid A)$.

59. According to recent research, air turbulence has increased in various regions around the world due to climate change. Turbulence makes flights bumpy and often delays the flights.

Assume that, an airplane observes severe turbulence, moderate turbulence or light turbulence with equal probabilities. Further, the chance of an airplane reaching late to the destination are 55%, 37% and 17% due to severe, moderate and light turbulence respectively.



On the basis of the above information, answer the following questions :

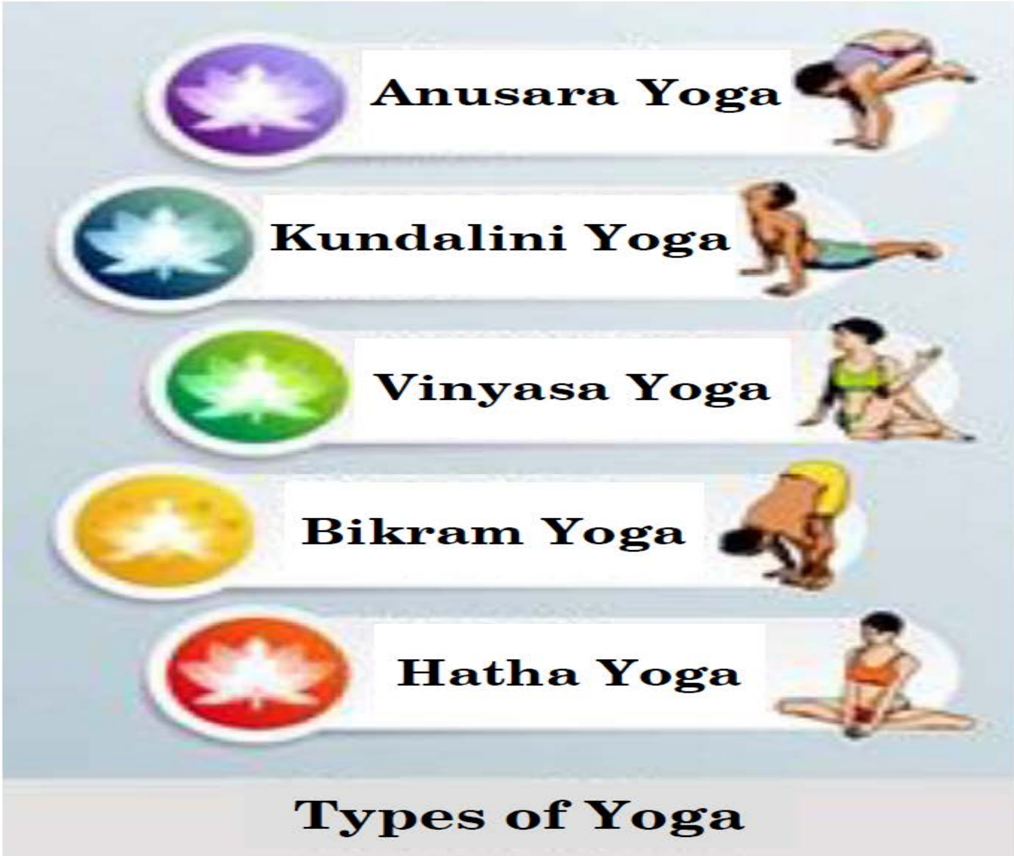
(i) Find the probability that an airplane reached its destination late. *2*

(ii) If the airplane reached its destination late, find the probability that it was due to moderate turbulence. *2*

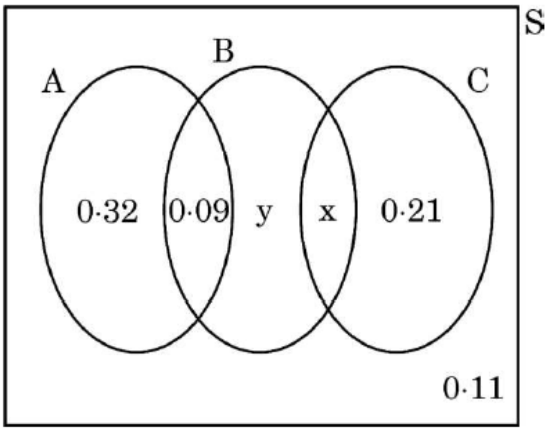
60.

Case Study – 1

There are different types of Yoga which involve the usage of different poses of Yoga Asanas, Meditation and Pranayam as shown in the figure below :



The Venn diagram below represents the probabilities of three different types of Yoga, A, B and C performed by the people of a society. Further, it is given that probability of a member performing type C Yoga is 0·44.



On the basis of the above information, answer the following questions :

(i) Find the value of x. *1*

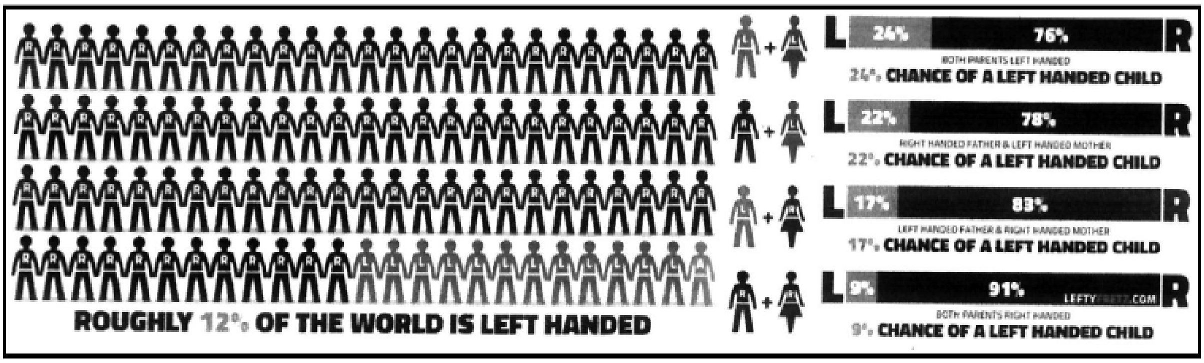
(ii) Find the value of y. *1*

(iii) (a) Find $P\left(\frac{C}{B}\right)$. *2*

OR

(iii) (b) Find the probability that a randomly selected person of the society does Yoga of type A or B but not C. *2*

61. Recent studies suggest that roughly 12% of the world population is left handed.



Depending upon the parents, the chances of having a left handed child are as follows :

A : When both father and mother are left handed :
Chances of left handed child is 24%.

B : When father is right handed and mother is left handed :
Chances of left handed child is 22%.

C : When father is left handed and mother is right handed :
Chances of left handed child is 17%.

D : When both father and mother are right handed :
Chances of left handed child is 9%.

Assuming that $P(A) = P(B) = P(C) = P(D) = \frac{1}{4}$ and L denotes the event that child is left handed.

Based on the above information, answer the following questions :

(i) Find $P(L/C)$

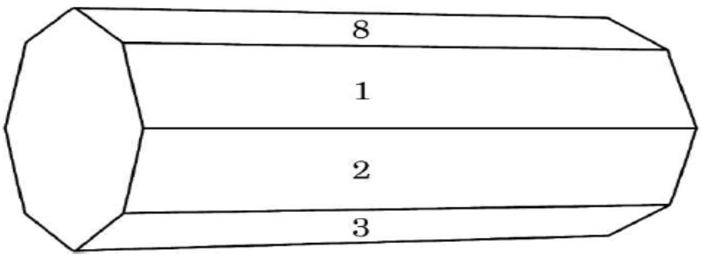
(ii) Find $P(\overline{L}/A)$

(iii) (a) Find $P(A/L)$

OR

(b) Find the probability that a randomly selected child is left handed given that exactly one of the parents is left handed.

62. An octagonal prism is a three-dimensional polyhedron bounded by two octagonal bases and eight rectangular side faces. It has 24 edges and 16 vertices.



The prism is rolled along the rectangular faces and number on the bottom face (touching the ground) is noted. Let X denote the number obtained on the bottom face and the following table give the probability distribution of X.

X :	1	2	3	4	5	6	7	8
P(X) :	p	2p	2p	p	2p	p^2	$2p^2$	$7p^2 + p$

Based on the above information, answer the following questions :

(i) Find the value of p.

(ii) Find $P(X > 6)$.

(iii) (a) Find $P(X = 3m)$, where m is a natural number.

OR

(iii) (b) Find the mean $E(X)$.

63. A building contractor undertakes a job to construct 4 flats on a plot along with parking area. Due to strike the probability of many construction workers not being present for the job is 0·65. The probability that many are not present and still the work gets completed on time is 0·35. The probability that work will be completed on time when all workers are present is 0·80.

Let : E_1 : represent the event when many workers were not present for the job;

E_2 : represent the event when all workers were present; and

E : represent completing the construction work on time.

Based on the above information, answer the following questions :

(i) What is the probability that all the workers are present for the job ?

(ii) What is the probability that construction will be completed on time ?

(iii) (a) What is the probability that many workers are not present given that the construction work is completed on time ?

OR

(iii) (b) What is the probability that all workers were present given that the construction job was completed on time ?

64. A salesman receives a commission for each sale he makes together with a fixed daily income. The number of sales he makes in a day along with their probabilities are given in the table below :

X :	0	1	2	3	4	5
P(X) :	0.42	3k	0.3	0.05	2k	0.03

His daily income Y (in ₹) is given by :

$$Y = 800X + 50$$

On the basis of the above information, answer the following questions :

(i) Find the value of k.

(ii) Evaluate $P(X \geq 3)$.

(iii) (a) Calculate the expected weekly income of the salesman assuming he works five days per week.

OR

(iii) (b) Calculate the expected weekly income of the salesman assuming he works only for three days of the week.

65.	In a group activity class, there are 10 students whose ages are 16, 17, 15, 14, 19, 17, 16, 19, 16 and 15 years. One student is selected at random such that each has equal chance of being chosen and age of the student is recorded.

On the basis of the above information, answer the following questions :

(i) Find the probability that the age of the selected student is a composite number.

(ii) Let X be the age of the selected student. What can be the value of X ?

(iii) (a) Find the probability distribution of random variable X and hence find the mean age.

OR

(iii) (b) A student was selected at random and his age was found to be greater than 15 years. Find the probability that his age is a prime number.

66.

A bank offers loan to its customers on different types of interest namely, fixed rate, floating rate and variable rate. From the past data with the bank, it is known that a customer avails loan on fixed rate, floating rate or variable rate with probabilities 10%, 20% and 70% respectively. A customer after availing loan can pay the loan or default on loan repayment. The bank data suggests that the probability that a person defaults on loan after availing it at fixed rate, floating rate and variable rate is 5%, 3% and 1% respectively.

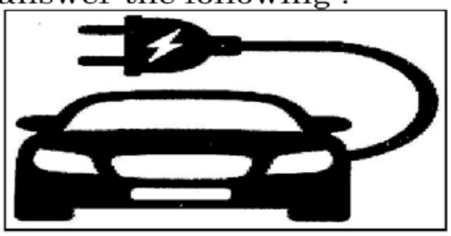
Based on the above information, answer the following :

(i) What is the probability that a customer after availing the loan will default on the loan repayment ? **2**

(ii) A customer after availing the loan, defaults on loan repayment. What is the probability that he availed the loan at a variable rate of interest ? **2**

67. Three persons viz. Amber, Bonzi and Comet are manufacturing cars which run on petrol and on battery as well. Their production share in the market is 60%, 30% and 10% respectively. Of their respective production capacities, 20%, 10% and 5% cars respectively are electric (or battery operated).

Based on the above, answer the following :



(i) (a) What is the probability that a randomly selected car is an electric car ? **2**

OR

(i) (b) What is the probability that a randomly selected car is a petrol car ? **2**

(ii) A car is selected at random and is found to be electric. What is the probability that it was manufactured by Comet ? **1**

(iii) A car is selected at random and is found to be electric. What is the probability that it was manufactured by Amber or Bonzi ? **1**

68.	Some students are having a misconception while comparing decimals. For example, a student may mention that $78 \cdot 56 > 78 \cdot 9$ as $7856 > 789$. In order to assess this concept, a decimal comparison test was administered to the students of class VI through the following question : In the recently held Sports Day in the school, 5 students participated in a javelin throw competition. The distances to which they have thrown the javelin are shown below in the table :

Name of student	Distance of javelin (in meters)
Ajay	47·7
Bijoy	47·07
Kartik	43·09
Dinesh	43·9
Devesh	45·2

The students were asked to identify who has thrown the javelin the farthest.

Based on the test attempted by the students, the teacher concludes that 40% of the students have the misconception in the concept of decimal comparison and the rest do not have the misconception. 80% of the students having misconception answered Bijoy as the correct answer in the paper. 90% of the students who are identified with not having misconception, did not answer Bijoy as their answer.

On the basis of the above information, answer the following questions :

(i) What is the probability of a student not having misconception but still answers Bijoy in the test ? *1*

(ii) What is the probability that a randomly selected student answers Bijoy as his answer in the test ? *1*

(iii) (a) What is the probability that a student who answered as Bijoy is having misconception ? *2*

OR

(iii) (b) What is the probability that a student who answered as Bijoy is amongst students who do not have the misconception ? *2*